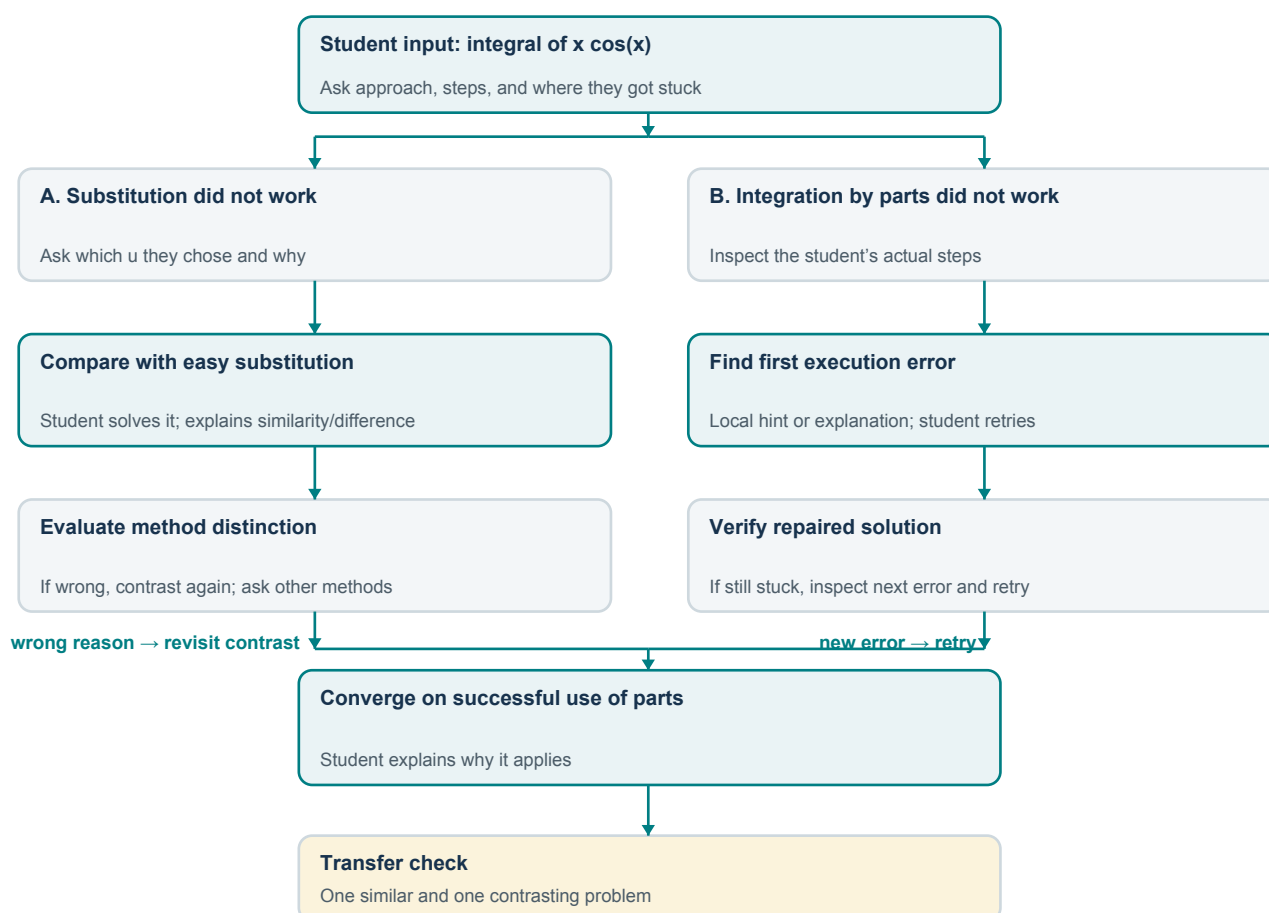


Digital Professor: A Concrete Integration Tutoring Workflow

A project-level system flow built from the team member's $\int x \cos(x) dx$ tutoring diagram. It specifies what each stage reads, decides, and passes onward.

Starting point: The tutor asks the student what they tried and where they got stuck. Two common replies lead to different support: “substitution did not work” and “integration by parts did not work.”

Figure 1. The instructional path translated into system branches



The flowchart preserves the human tutor's structure: elicit → distinguish approaches → address the specific difficulty → return the problem to the student → check transfer. A request for missing steps can go straight back to the student; detailed course retrieval is used when a method, example, or mathematical claim must be supported.

1 Receive input and route the student's attempt

The first system task is to turn an informal student message into a faithful record. For the pilot, accept typed mathematics and text. The system preserves the exact wording and asks for missing working rather than treating “did not work” as a complete diagnosis.

Suggested method

A schema-constrained LLM extraction reads the student's language and proposes fields. Deterministic checks verify required fields, course IDs, and allowed route labels. A separate trained NLP classifier is not needed for this small set of branches; compare the extraction with a simple rule baseline during evaluation. A model's self-reported confidence is not used as proof of understanding.

Structured output: StudentInput

Field	What to capture	Example
problem_id	Normalized course problem plus verbatim problem text.	calc1_integral_x_cos_x
student_goal	Diagnose my work, explain method, get a hint, or unspecified.	Diagnose why my method failed.
attempted_method	substitution, integration by parts, other, or unknown.	substitution
attempted_steps	Ordered steps with the student's exact symbols.	“I chose $u = x$ ” (if supplied).
method_reason	Why the student chose this method; unknown if unstated.	unknown
stuck_point	Specific line or operation where progress stopped.	Cannot express the remaining integral in u .
prior_help	Hints or examples already given in this session.	none
topic_candidates	Course/chapter IDs and concepts for possible retrieval.	Calculus I / integration methods.
retrieval_request	Whether detailed material is needed now; if yes, what to seek.	No: first ask for the missing setup.
missing_information	What would change the next pedagogical choice.	Choice of u and reason for choosing it.
input_state	ready, needs_steps, ambiguous_method, unreadable, out_of_scope.	needs_steps

Route definitions

Input state	Definition and immediate action
ready	Problem and relevant attempted steps are legible. Route to the method branch; retrieve detailed material when the next action needs it.
needs_steps	A method is named but no actual working is shown. Ask for u/du or the integration-by-parts setup.
ambiguous_method	“I tried something” or multiple methods without a clear order. Ask which attempt to inspect first.
unreadable / out_of_scope	Request a clearer representation or state the course boundary; do not invent a diagnosis.

Important: “substitution did not work” is evidence about the attempted method, not yet evidence that the student misunderstands substitution.

2 Retrieve the course evidence needed for this branch

Detailed retrieval is conditional. When the immediate action is “show me your u and dv,” the system can ask without fetching a lecture passage. Retrieve when it must verify a method or calculation, select a course example, explain a contrast, or create a transfer task. Retrieval supplies mathematically authoritative content; it does not prove a student misconception.

Course preparation

Ovunc's course component supplies approved material organized by objective, prerequisite, method, worked example, and exercise. Keep formulas with their worked steps. Record source type and version: syllabus for scope, verified teaching material for classroom claims, and textbook or instructor-approved material for method criteria and explanations.

Retrieval method and gate

For the first small course packet, use a curated mapping from the problem and attempted method to source units. As materials grow, filter by course/topic, search method names and mathematical structure, then compare keyword retrieval with semantic or hybrid retrieval on labeled tutoring questions. Use a reranker only if it improves the evidence set. Tune relevance thresholds and top-k on examples; do not choose them by convention.

Student branch	Required evidence	Why it is needed
Substitution attempt	Substitution criterion; one easy valid substitution example; why the current integrand does not fit that pattern directly; integration-by-parts criterion.	Support comparison and method choice without simply announcing the answer.
Integration-by-parts attempt	Formula; worked setup of u and dv; derivative and antiderivative checks; expected result or reference solution.	Locate the first actual execution error and validate a repaired step.
After success	One similar integration-by-parts problem and one contrasting problem using another method.	Test whether the student can select a method beyond the original exercise.

Structured output: EvidenceBundle

Field	Required content
objective_id / source_version	Stable learning objective and versioned source context.
retrieved_units	For each unit: source ID, section, excerpt, source authority, and relevance reason.
method_criteria	Conditions that distinguish substitution from integration by parts.
reference_steps	Verified intermediate steps; available to validation, not automatically shown to the student.
comparison_item / transfer_items	Easy substitution contrast; similar and different follow-up problems.
course_evidence_state	supported, partial, missing, or conflict; with a specific reason.
student_evidence_state	enough or insufficient to infer the student's difficulty; independent of course evidence.

Continue only if evidence supports the intended teaching action. If the comparison example or reference solution is missing, retrieve again or request human review rather than inventing one.

3 Diagnose the student's state from actual work

The system compares the student's ordered steps with the reference method and maintains two separate layers: observed facts and possible explanations. "The student chose $u = x$ " is observed; "the student believes every product calls for substitution" requires their stated reasoning or further probing.

Branch A: substitution did not work

Inspect the chosen u , computed du , substitution into the integral, and the student's reason for selecting the method. If the student can already explain why the easy substitution example differs from $\int x \cos(x) dx$, they may understand substitution; the next need is choosing another method. If they incorrectly claim the two problems have the same structure, the misconception is supported and a contrast is appropriate.

Branch B: integration by parts did not work

Ask for the actual u and dv , derivatives/antiderivatives, substitution into the integration-by-parts formula, simplification, and final result. Mark the first incorrect transition. Possible error categories include setup, derivative, antiderivative, sign in the formula, or later algebra. The method choice is sound; the intervention targets the first error.

Structured output: StudentState

Field	Definition
observed_approach	Method attempted and steps actually supplied.
correct_components	Valid steps or method choices to preserve.
first_meaningful_error	Earliest wrong transition; unknown when steps are missing.
suspected_cause	Execution slip, method-selection misconception, missing prerequisite, or unknown.
evidence_for_cause	Student quote/step and course criterion that support the hypothesis.
alternative_explanations	Other plausible reasons for the same visible error.
prior_interventions	Examples, hints, explanations, and student responses already tried.
understanding_evidence	Recognition, assisted execution, independent execution, explanation, transfer.
diagnosis_state	supported, ambiguous, insufficient, or conflicting.

Diagnosis-state definitions

State	Operational meaning	Next action
supported	A specific first error or misconception is grounded in steps and stated reasoning.	Targeted intervention.
ambiguous	At least two causes fit the visible work.	Ask a discriminating question.
insufficient	The setup or explanation needed to distinguish causes is absent.	Ask for the missing step.
conflicting	Student steps, reference source, or independent checks disagree.	Clarify or human review.

4 Choose the pedagogical action and response

A decision layer selects the next tutoring action from the student state, then asks a model to phrase that action. The action record names the exact gap being addressed, the amount of help allowed, and what new student evidence should appear next.

Model interface for the first pilot

Use two logical model tasks per turn, which can use the same model: first, interpret the new student input and propose a state update and retrieval request; second, after any needed retrieval, propose the teaching action and message. The system checks both structured results. If evaluation later shows that action selection is sound but wording fails, split those duties further; if simple turns can be handled with fewer calls, avoid extra latency.

Observed condition	Selected action	Student-facing example
Substitution named; no setup supplied	inspect / elicit	"What did you choose for u, and what is du?"
Student cannot solve an easy substitution example	review prerequisite	"Try this simpler substitution first; show your choice of u."
Student says current integral and easy substitution have the same structure, for an incorrect reason	contrast / explain	"Where is the derivative of your chosen inner expression in each integrand?"
Student correctly explains the difference but has no next method	probe method choice	"What other integration techniques have you seen that could apply here?"
Student names parts among several methods	compare options	"Why would parts help here, and why do the other methods fit less well?"
Parts selected; first execution error visible	local hint / retry	"Check the sign when applying the formula to that line. Try it again."
Student solves original after help	check understanding	"Why does choosing parts make this integral simpler?"
Student explains independently	transfer	Give one similar and one different integral; ask which method fits and why.

Structured output: DecisionRecord

Field	Definition
action	One permitted tutor action: elicit, inspect, probe, hint, contrast, explain, demonstrate, retry, verify, check understanding, or transfer.
target_gap / reason	Which observed student evidence makes this the next useful action.
help_level	Question, directional hint, explicit explanation, or demonstration.
expected_next_evidence	What the next student reply must show to advance the state.
alternative_route	What to do if the student remains stuck or answers differently.

The system gives one useful intervention at a time. If a hint fails, it changes the action or help level; it does not repeat the same hint in new words.

5 Validate the response and repeat the tutoring loop

Before the student sees a message, the system checks the evidence, mathematics, and teaching strategy. It then waits for a student reply and re-enters the same input → optional retrieval → diagnosis → decision loop with the prior student state attached.

Gate	Pass criterion	Failure branch
Structure	Input, state, evidence, and decision fields are present and use allowed values.	Repair once; hold output if still invalid.
Source grounding	Claims about method criteria and course coverage trace to approved material.	Retrieve again or remove claim.
Mathematical correctness	The comparison, hint, and reference solution are correct.	Independent check or human review.
Pedagogical fit	The message matches the selected action and addresses the observed difficulty.	Choose a different action or regenerate.
Disclosure control	A probe or hint does not reveal the full method or result too early.	Rewrite at the same help level.
State consistency	The new inference does not contradict observed student work or past responses.	Preserve uncertainty; ask for clarification.

How state changes after each reply

Record what changed, how much help the student had, whether the original difficulty remains, and whether the response supports or refutes the previous diagnosis. A correct result after substantial prompting is “assisted execution.” Independent solving, a clear explanation of method choice, and selection on new problems are stronger evidence. The state estimate may be revised when the student’s reasoning contradicts the initial hypothesis.

Convergence and stopping conditions

Both original branches converge when the student can use integration by parts to solve the problem and explain why it fits. The member’s workflow then gives two further examples, one similar and one different. The system closes when the student appropriately chooses and explains methods on those examples, or when the student stops. Repeated failed interventions, unresolved source conflicts, or uncertain mathematics lead to a human tutor rather than an endless loop.

Understanding evidence

Level	What the student has actually demonstrated
0–1 Limited / recognition	May name a method but cannot yet use or justify it.
2 Assisted execution	Completes a step after a hint or example.
3 Independent execution	Solves a comparable problem without the same help.
4 Explanation	Explains why parts applies and why substitution does not fit this structure directly.
5 Transfer	Chooses methods correctly for both similar and contrasting problems.

6 One concrete multi-turn trace

The following trace follows the substitution branch of the team member's diagram. Its student replies are illustrative; they are not outcomes of an implemented multi-turn system.

Turn	Evidence and state update	Action
1. Student: "I tried substitution; it did not work."	Approach is known; setup and reason are absent. Diagnosis state = insufficient.	Ask for choice of u , du , and the exact stuck step.
2. Student: "I chose $u = \cos(x)$, but du has $\sin(x)$, so I could not replace $x \, dx$."	Student has found a mismatch. A method-selection misconception is possible but unproven. Diagnosis state = ambiguous.	Give an easy valid substitution example; ask the student to solve and compare structures.
3. Student: "The example has an inner derivative beside the function; my original integral does not."	Correct distinction is evidenced. The student understands why this substitution setup stalls.	Ask which other techniques could apply and why.
4. Student names integration by parts.	Method recognition is evidenced; execution remains untested.	Ask the student to set up u and dv and walk through the solution.
5. Student makes a sign error in the parts formula.	Method choice remains correct. First execution error is localized to the sign.	Give a local hint on that line; have the student retry.
6. Student solves and explains the choice.	Assisted execution and explanation are observed. Transfer is not yet known.	Present $\int x e^x \, dx$ (similar: parts) and $\int 2x \cos(x^2) \, dx$ (contrast: substitution); ask method and reason.

Alternative path from the same initial question

If the student initially says "integration by parts did not work," the system skips the easy substitution comparison. It asks for the worked solution, identifies the first wrong step, gives a local correction, and returns the problem to the student. If the student cannot repair the step, the system increases help or asks a human to review. The two paths meet at explanation and transfer.

7 One complete retrieval / model handoff

This example makes the interfaces inspectable. It is a proposed mock interaction; the source IDs are illustrative catalog IDs that Ovunc would replace with approved course-unit identifiers. The student says: “For $\int x \cos(x) dx$ I chose $u = \sin(x)$. I still have x , so I am stuck.”

Handoff	Structured result in this turn	Decision it enables
Jonathan → Sean: raw evidence	problem = $\int x \cos(x) dx$; attempt = $u = \sin(x)$; stuck point = x remains; reason for method choice = absent.	Preserve the student’s exact words and steps.
Model task 1: extraction	attempted_method = substitution; observed_steps = [$u = \sin(x)$]; input_state = ready; diagnosis = uncertain; missing = “why this u ?”; retrieval_request = compare substitution and parts.	Ask a discriminating question or retrieve an approved comparison example.
Sean → Ovunc catalog: retrieval request	course = Calculus I; topic = integration methods; need = substitution criterion, simple valid example, parts criterion.	Return relevant course units only.
Retrieval result: EvidenceBundle	illustrative IDs: SUB-CRITERION-01, SUB-EXAMPLE-01, PARTS-CRITERION-01; each carries source type, location, version, excerpt; course_evidence_state = supported.	Supports a comparison without claiming the student has a misconception.
Model task 2: state and action	student_evidence_state = ambiguous; observed fact = x remains after substitution; hypothesis = method-choice confusion; action = contrast; expected evidence = explanation of why the valid substitution example differs.	Generate one comparison question.
Response gate and delivery	Check source IDs, mathematical truth, and help level; send: “Compare a simple valid substitution with this integral. What becomes du in each case, and what is left over?”	Student replies; the loop updates the state.
Persistent turn record	Save raw input, structured state version, retrieved source IDs/versions, chosen action, actual message, student response, latency, and cost.	Next turn can continue without reinterpreting the whole history.

Two sufficiency judgments

Student evidence sufficient? Can we infer why this student is stuck? Here, no: the chosen substitution is visible, but the reason is missing. Course evidence sufficient? Can approved material support the intended comparison? In this mock, yes. These judgments can differ; the system can ask a useful comparison question without pretending to know the student’s misconception.

If Ovunc’s approved source units are not available, the retrieval result must be “missing” rather than silently converting this mock example into a claimed course citation.

8 Evaluation and current project boundary

Use the team member's diagram as the initial reference trajectory, then add variations: correct substitution reasoning, incorrect comparison, wrong method choice, correct parts setup with a sign error, repeated failed hints, and a student who gets the result without understanding why. Human tutors label acceptable next actions; the system need not copy the exact wording or sequence when another path is educationally sound.

Stage	What to evaluate	Evidence
Input extraction	Did it preserve the actual method, steps, and uncertainty?	Field-level accuracy on labeled student messages.
Retrieval	Did it fetch the correct criteria, worked step, and contrast example?	Human relevance labels; recall@k and precision@k.
Diagnosis	Did it find the first error and avoid inventing a misconception?	Tutor-labeled error and ambiguity agreement.
Action	Was the next move appropriate for this branch and level of help?	Tutor agreement; unnecessary solution disclosure rate.
Intervention	Was the prompt mathematically correct, clear, and grounded?	Expert review and student-response usefulness.
State update	Did it distinguish recognition, assisted work, explanation, and transfer?	Turn-by-turn human state labels.
Session	Did the student choose and apply methods on new cases?	Transfer outcomes, number of turns, time, cost, human handoffs.

Team interfaces

Jonathan supplies the problem, raw steps, and each follow-up response. Ovunc supplies objective IDs and versioned course units. Yining's tutor-experience research informs the action vocabulary and help levels. Sean defines the model and retrieval interfaces, structured state handoffs, source tracing, validation, persistence, and per-turn time/cost recording. The pedagogical policy itself requires team agreement.

What exists today

The Week 2 prototype demonstrates one curated Chain Rule lesson packet, routing, model API calls, structured JSON, and basic local checks. The integration tutoring workflow in this report is a proposed next scenario and system design. Its persistent state, multi-turn routing, method-specific retrieval, mathematical verification, transfer check, and human handoff are specified here but have not been implemented.

Design choices to test before adding complexity

Start with curated course packets and schema-constrained extraction. Compare those baselines with larger retrieval methods or extra model calls only when turn-level evaluations show an improvement worth the latency and cost. Do not rely on model self-confidence, a single correct worked answer, or one retrieval score as proof of student understanding.

Sources

Project basis: Digital_Professor_Integration_Ideas (1).pdf (integration tutoring flow); Digital_Professor_Integration_Ideas_main.pdf (general tutoring loop); Week 2 prototype files. Technical background: [OpenAI Structured Outputs](#); [Microsoft retrieval evaluation](#); [OpenAI evaluation guidance](#); [Anthropic workflow guidance](#).